

In Silico Study On Dermal Interaction Of Bioactive Compound

CHAPTER 1

INTRODUCTION

1.1 Skin Structure and Physiology

The skin is the largest organ of the human body and acts as a protective barrier against external environmental conditions. It helps regulate body temperature, prevents excessive water loss, supports immune functions, and enables sensory responses such as touch and pressure. Due to its layered structure, the skin also controls the movement and absorption of substances applied to its surface, including drugs and cosmetic compounds. The skin mainly consists of three layers: epidermis, dermis, and hypodermis.

Epidermis

The epidermis is the outermost layer of the skin and protects the body from environmental exposure. It is mainly composed of keratinocytes and acts as the primary barrier controlling the entry of external substances. The stratum corneum, present within the epidermis, plays a major role in skin permeability and diffusion.

Dermis

The dermis lies beneath the epidermis and provides structural support to the skin through collagen and elastin fibers. It contains blood vessels, nerves, sweat glands, and hair follicles, and is involved in temperature regulation, nutrient supply, and sensory functions.

Hypodermis

The hypodermis is the deepest layer of the skin and mainly consists of adipose tissue and connective tissue. It provides cushioning, insulation, and energy storage while connecting the skin to underlying muscles and tissues.

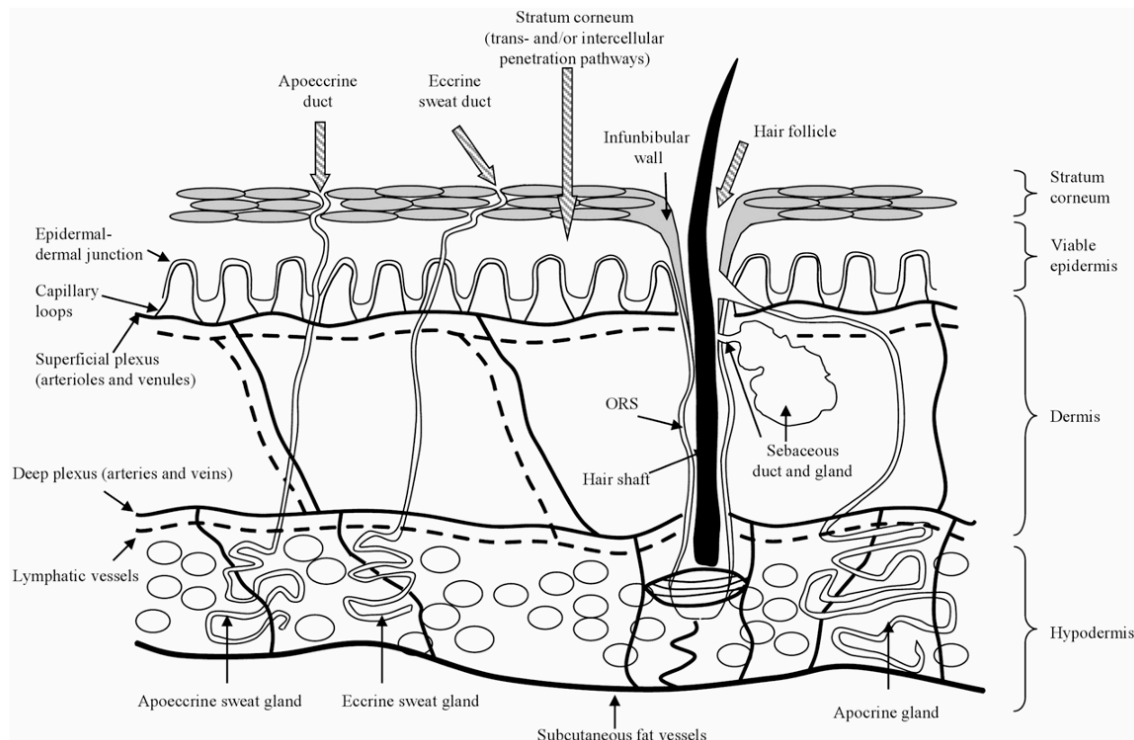


Figure 1: Cross-sectional structure of human skin.

Stratum Corneum as a Permeability Barrier

The stratum corneum is the outermost region of the epidermis and serves as the main barrier of the skin. It consists of dead corneocyte cells embedded within a lipid matrix, often described using the “brick-and-mortar” structure. This arrangement helps prevent excessive water loss and restricts the entry of external substances.

Skin permeability is strongly influenced by factors such as molecular weight, lipophilicity, polarity, and hydrogen bonding ability. Due to its lipid-rich nature, the stratum corneum limits the movement of large and highly water-soluble molecules, making it an important barrier in dermal diffusion and transdermal transport studies.

Functions of Skin

The skin is an important organ that performs many functions necessary for protecting and maintaining the body :

- It acts as a barrier against harmful microorganisms, chemicals, and ultraviolet rays from the environment.
- The skin also helps retain moisture by preventing excessive water loss and supports body temperature regulation through sweating and changes in blood flow.

- Enables sensations such as touch, pain, and temperature, allowing the body to respond to surrounding conditions.
- In addition, the skin contributes to immune protection and stores fat that can serve as an energy source.
- It also plays a role in the production of vitamin D.

Because the skin controls the movement of substances into the body, it is widely studied in cosmetic and pharmaceutical research. Understanding how compounds move through different skin layers is important for analyzing the dermal behavior of rice-derived bioactive compounds used in topical formulations.

1.2 Transdermal and Topical Drug Delivery

Concept of Topical Delivery

Topical delivery involves applying medicinal or cosmetic formulations directly onto the skin to produce localized effects. Products such as creams, gels, lotions, ointments, and serums mainly act within the outer skin layers rather than entering the bloodstream. This method is widely used in skincare and dermatological treatments because it allows active ingredients to interact directly with the skin. Rice-derived bioactive compounds are commonly included in such formulations due to their antioxidant and skin-protective properties.

Transdermal Drug Delivery

In transdermal delivery, compounds are designed to pass through the skin and enter the bloodstream to produce systemic effects. The movement of molecules across skin layers depends on factors such as molecular size, polarity, solubility, and lipophilicity. Since experimental studies can be time-consuming, computational approaches are often used to study and predict diffusion behavior through the skin.

Advantages of Topical Delivery

Topical delivery provides localized action, controlled release, ease of application, and reduced systemic side effects. It is especially useful in cosmetic formulations because bioactive compounds can directly interact with skin tissues and provide antioxidant, anti-inflammatory, and anti-aging benefits.

1.3 Skin Permeability and logKp

Skin permeability refers to the ability of a substance to move from the skin surface into deeper layers. This property is important in skincare and cosmetic formulations because compounds must cross the skin barrier to produce their intended effects. The stratum corneum acts as the primary barrier controlling the entry of external molecules.

The movement of compounds through the skin depends on several molecular properties, including molecular size, lipophilicity, polarity, hydrogen bonding ability, and solubility. Smaller molecules with balanced lipid and water affinity generally show better permeability.

Skin permeability is commonly represented using the permeability coefficient (K_p), expressed as:

$$K_p = J/C$$

where J represents steady-state flux and C represents the concentration of the compound. The logarithmic form, $\log K_p$, is widely used to compare permeability values. Compounds with values closer to zero generally show better skin permeation, whereas more negative values indicate lower permeability.

Important physicochemical descriptors influencing skin permeability include molecular weight, LogP , topological polar surface area (TPSA), hydrogen bonding capacity, and solubility. Understanding these properties helps predict the dermal diffusion behavior of rice-derived bioactive compounds used in topical formulations.

1.4 Rice-Based Bioactive Compounds

Rice-derived bioactive compounds are widely used in cosmetic and skincare research because of their antioxidant and skin-protective properties. Rice bran, pigmented rice, and fermented rice contain several phytochemicals that help reduce oxidative stress and support skin health. Due to their natural origin and compatibility with different skin types, these compounds are commonly incorporated into topical formulations.

Important rice-derived bioactive compounds include:

- Ferulic acid
- Caffeic acid
- γ -Oryzanol
- p-Coumaric acid
- Vanillic acid
- Chlorogenic acid
- Anthocyanins

These compounds are known for their antioxidant, anti-inflammatory, and anti-aging activities. They are commonly used in creams, serums, moisturizers, and other skincare products. In addition to cosmetic applications, some rice bioactives are also associated with skin repair and protection against environmental stress, making them suitable for diffusion and permeability studies using computational approaches.

1.5 Need for Computational Prediction

Experimental techniques such as in vitro diffusion and ex vivo skin permeation studies are commonly used to measure skin permeability, but they are often time-consuming, expensive, and require specialized facilities. Because of these limitations, computational methods are widely used for preliminary screening of compounds.

Approaches such as QSAR and machine learning can predict skin permeability using molecular descriptors including molecular weight, LogP, TPSA, and hydrogen-bonding properties. These methods help identify compounds with favorable dermal diffusion characteristics while reducing the need for extensive experimental studies. In this study, computational prediction methods were applied to evaluate rice-derived bioactive compounds for topical and cosmetic applications.

1.6 Virtual Cell (VCell) Modeling

Virtual Cell (VCell) is a computational platform used to simulate biological processes such as diffusion and molecular transport. It helps predict how compounds move across different compartments over time using mathematical models.

In this study, VCell was used to develop a multilayer skin model consisting of the vehicle, lipid layer, corneocyte layer, viable epidermis, and dermis. Appropriate diffusion coefficients

and layer properties were assigned to study the movement of rice-derived bioactive compounds through the skin.

The simulations generated concentration gradients and diffusion profiles that helped analyze compound penetration and distribution across skin layers. These results complemented the machine learning analysis and supported the overall computational workflow of the study.

1.7 Machine Learning in Permeability Prediction

Machine learning is increasingly being used in pharmaceutical and cosmetic research to predict important properties of compounds in a faster and more efficient manner. In skin permeability studies, these methods help identify relationships between molecular descriptors and permeability behavior, making it easier to screen compounds with suitable dermal absorption properties.

Descriptors such as molecular weight, LogP, TPSA, hydrogen bonding properties, and solubility are commonly used as input features for predicting skin permeability (logKp). Feature selection also plays an important role in identifying descriptors that contribute significantly to prediction performance.

Several regression models, including Linear Regression, Random Forest Regression, Gradient Boosting, and Support Vector Regression (SVR), are widely used in permeability prediction studies to analyze the relationship between molecular properties and dermal absorption behavior.

1.8 Research Gap

Despite the advances made in skin permeability research, most studies rely on either simulation methods or machine learning models alone. Only a limited number of studies have combined both approaches to obtain a more complete understanding of dermal transport.

Research on rice-derived bioactive compounds is also relatively scarce, even though these natural molecules are increasingly used in cosmetic and skincare products. Most existing studies have focused on synthetic pharmaceutical compounds rather than plant-based ingredients.

This highlights the need for an integrated computational approach that combines molecular descriptors, machine learning, and biological simulation to efficiently identify rice-derived compounds with strong potential for topical and cosmetic applications.

CHAPTER 2

OBJECTIVE

The present study was carried out to understand the dermal permeation behavior of rice-derived bioactive compounds and to identify compounds suitable for topical and cosmetic applications. Previous studies have shown that computational methods, molecular descriptors, and diffusion simulations can effectively predict skin permeability and analyze compound transport across skin layers.

Computational approaches such as QSAR, machine learning, and diffusion-based simulations are widely used because they reduce the need for extensive experimental studies. Molecular properties including molecular weight, lipophilicity, and topological polar surface area strongly influence dermal absorption behavior. Combining these methods provides a simple and efficient approach for studying compound diffusion and permeability through the skin.

CHAPTER 3

METHODOLOGY

This study used a computational approach to evaluate the skin permeability of rice-derived bioactive compounds. The methodology included molecular descriptor collection, data preprocessing, machine learning-based prediction, and diffusion simulation using VCell. Similar computational methods are commonly used in skin permeability and QSAR studies to predict logK_p and analyze compound transport across skin layers.

3.1 Overall Workflow

The workflow of the study involved dataset collection, descriptor extraction, preprocessing, machine learning model development, VCell simulation, and validation. Rice-derived bioactive compounds were first selected from literature, and their physicochemical properties

were collected from public databases. After preprocessing the dataset, machine learning models were developed to predict skin permeability. VCell simulations were then carried out to study diffusion across different skin layers, and the obtained results were compared with literature-reported values for validation.

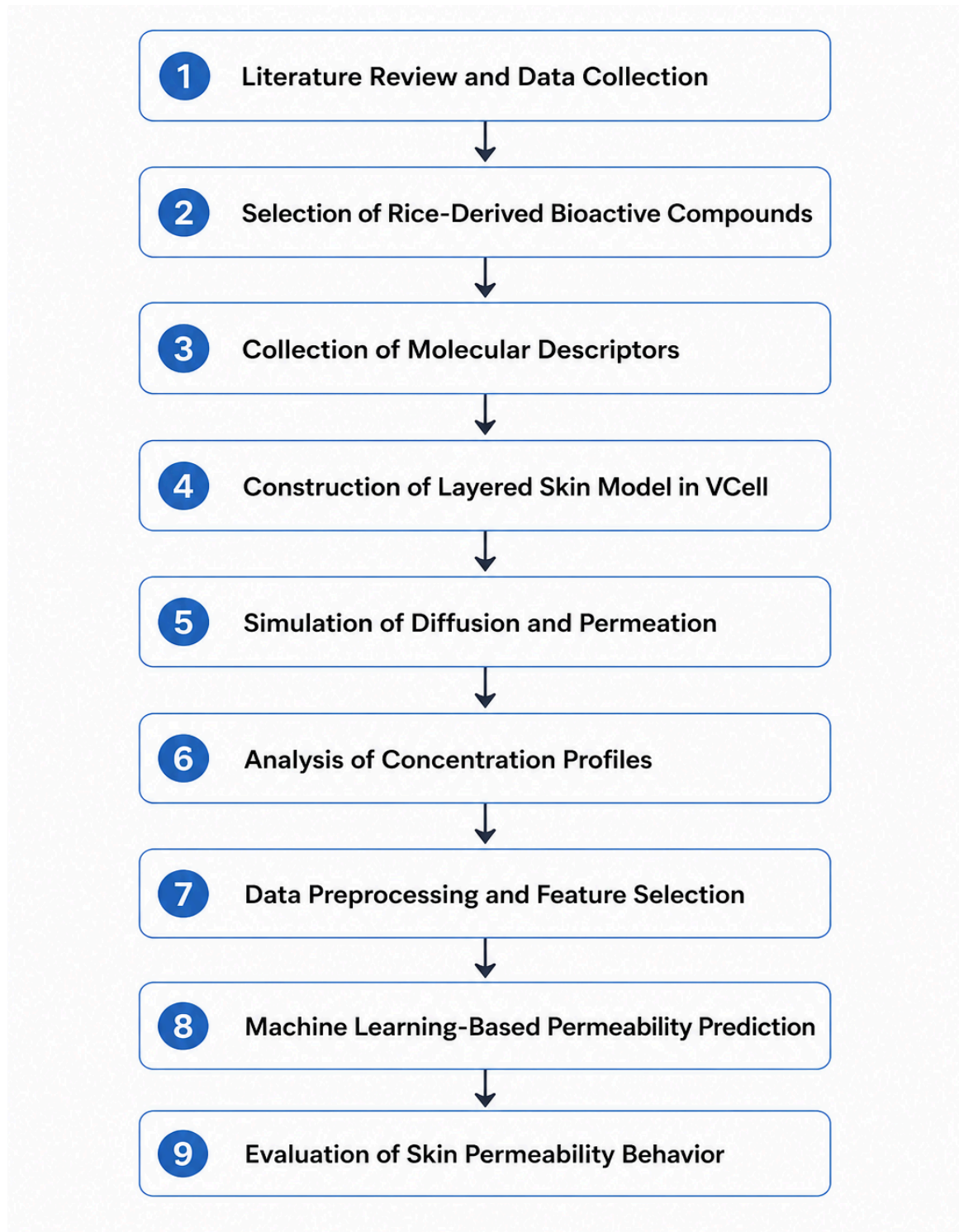


Figure 2: Workflow of the study.

3.2 Dataset Collection

A dataset containing nearly 130 rice-derived bioactive compounds was prepared for this study. The compounds were selected from published literature related to rice bran, pigmented rice, and fermented rice. Physicochemical information for these compounds was collected from databases such as PubChem, DrugBank, and other scientific sources.

The collected descriptors included molecular weight, XlogP, topological polar surface area (TPSA), hydrogen bond donors and acceptors, rotatable bonds, aqueous solubility (LogS), and skin permeability (logKp). These properties were chosen because they are known to influence dermal absorption and skin permeability behavior.

Table 1: Molecular descriptor data collected for selected rice-derived (130) compounds.

Bioactive compound	Mol.Weight(g/mol)	TPSA (Å ²)	LogP	LogD	Rotatable Bonds
Ferulic acid	194.06	66.76	1.344	1.241	3
Caffeic acid	180.16	77.76	1.01	0.531	2
p-Coumaric acid	164.05	57.53	1.438	1.05	2
Sinapic / Sinapinic acid	224.07	75.99	1.52	2.348	4
Protocatechuic acid	154.03	77.76	1.041	1.003	1
γ-Oryzanol	602.43	55.76	9.366	6.32	9
Vanillic acid	168.04	66.76	1.396	3.428	2
Chlorogenic acid	354.31	164.75	-0.085	-0.351	5
Gallic acid	170.02	97.99	0.645	0.343	1
Phytic acid	659.86	400.56	-1.7	-0.227	12

3.3 Data Preprocessing

The collected dataset was checked for missing values, duplicate entries, and inconsistencies to improve data quality. Descriptors with large amounts of missing information were removed, and duplicate compounds were excluded from the dataset. The physicochemical properties were also standardized where required to maintain consistent numerical values for model development.

Correlation analysis was carried out to study relationships between descriptors and identify redundant variables. Based on the analysis, descriptors such as pKa and melting point were

removed because of incomplete data and lower relevance to model performance. The final dataset contained descriptors with adequate data availability and importance for skin permeability analysis.

3.4 Governing Equations

Skin permeability was represented using the permeability coefficient (Kp), which describes the rate at which a compound moves through the skin. The equation used to calculate is:

$$\text{Logkp} = -3.2 + \log\text{Kp_FA} + 0.08*\text{LogP} + 0.015*(\text{LogP}^2) - 0.0005*\text{TPSA} - 0.0008*\text{MW} + 0.05*(y/200) + 0.00005*t.$$

Molecular properties strongly influence skin permeability. In general, compounds with higher molecular weight show lower permeability, while moderate LogP values support diffusion through the lipid-rich stratum corneum. Similarly, higher TPSA values often reduce permeability because highly polar molecules move less efficiently through hydrophobic skin layers.

3.5 VCell Model Development

3.5.1 Geometry and Compartments

A multilayer skin model was developed in VCell to study the diffusion of rice-derived bioactive compounds. The model included five compartments: vehicle, lipid layer, corneocyte layer, viable epidermis, and dermis. This arrangement was designed based on the layered structure and barrier properties of human skin reported in literature.

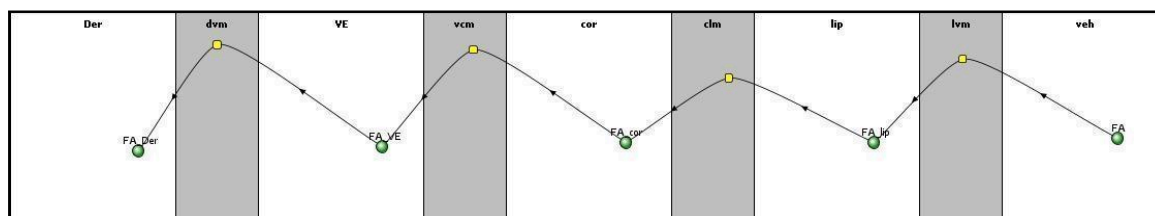


Figure 3: Multilayer skin model constructed in VCell for diffusion simulation.

3.5.2 Simulation Parameters

An initial concentration was introduced in the vehicle compartment to represent topical application of compounds. Different diffusion coefficients were assigned to each skin layer based on literature-reported values. Suitable boundary conditions were applied to allow concentration-driven transport between adjacent layers. Simulations were carried out over a fixed time period to observe concentration changes and diffusion behavior across the skin model.

3.5.3 VCML Model Setup

The diffusion model was developed using Virtual Cell Markup Language (VCML) to study the movement of compounds across different skin layers. The simulations generated outputs such as concentration changes over time and diffusion patterns within the model. These results were later exported for visualization and analysis.

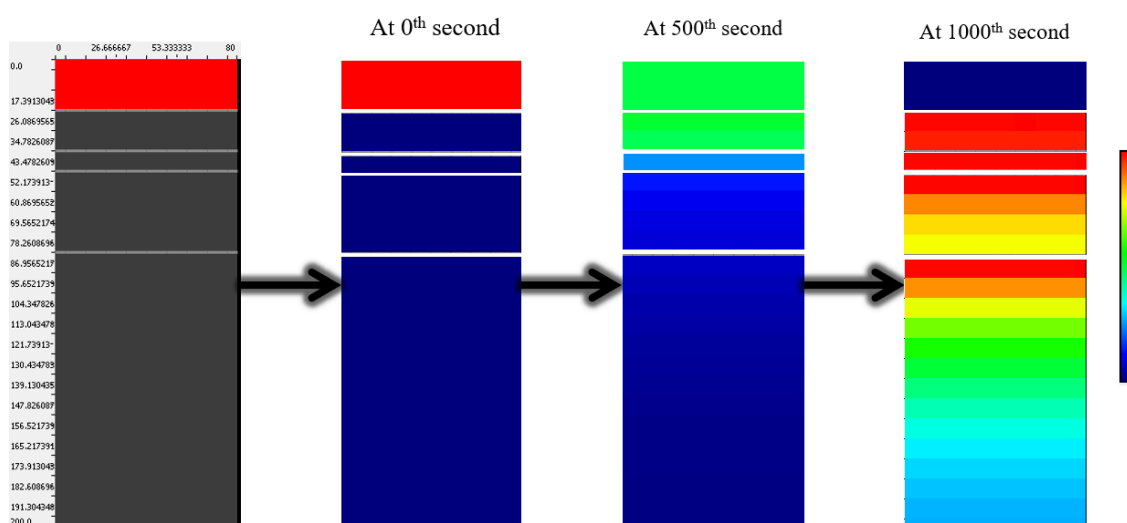


Figure 4: Diffusion profile of compounds across different skin layers in the VCell model over time.

3.6 Machine Learning Methodology

3.6.1 Data Preparation for Machine Learning

For machine learning analysis, the collected dataset was organized into two portions so that one part could be used for model learning and the other for prediction assessment. Numerical descriptor values were scaled to avoid large variations between features and to support stable model behavior. The selected molecular properties acted as predictor variables, while logK_p values were used as the output variable during analysis.

3.6.2 Descriptor Selection

To understand which molecular properties contribute more strongly to skin permeability, descriptor selection was carried out during analysis. Properties such as molecular size, lipophilicity, and polar surface area showed significant influence on permeability behavior. Additional descriptors related to hydrogen bonding, bond flexibility, and solubility were also included to provide broader chemical information for prediction. Interaction-based features derived from selected descriptors were further incorporated during model development.

3.6.3 Regression-Based Prediction Models

Multiple regression algorithms were explored to estimate skin permeability from molecular descriptor data. Linear Regression was used to observe basic linear trends between descriptors and permeability values. More advanced ensemble approaches, including Random Forest Regression and Gradient Boosting, were applied to capture complex relationships present in the dataset. Support Vector Regression was additionally used to study nonlinear variations in permeability behavior.

3.6.4 Performance Analysis

The developed models were examined using statistical evaluation measures such as R^2 score, root mean square error (RMSE), and mean absolute error (MAE). These parameters were used to compare predicted permeability values with observed values and estimate the accuracy of each model. Graphical comparisons between predicted and actual outputs were also generated for visualization of model behavior.

3.6.5 Reliability Assessment Using Cross-Validation

To examine the consistency of the developed models, a five-fold cross-validation procedure was performed. In this method, different portions of the dataset were repeatedly used for training and testing. This approach helped provide a more dependable estimate of model stability and reduced the possibility of overfitting during prediction.

3.7 Validation Strategy

The predicted permeability trends and diffusion behavior obtained from computational analysis were compared with previously reported studies available in literature. Agreement

between the predicted observations and reported permeability trends was considered supportive of model reliability.

Experimental logKp values reported by Cheruvu et al. (2022) were used as reference data for validation. The dataset contained compounds with varied physicochemical properties and permeability values mainly corresponding to the stratum corneum. Predicted permeability values obtained from the model were compared with these experimental values to evaluate model performance and reliability.

Table 2: Comparison of experimental and predicted logKp values for selected compounds.

Compounds	Expected value	Obtained value	Difference
Raffinose	-4.3	-4.53	-0.23
Linoleic acid	-2.43	-2.7	-0.27
Oleic acid	-2.89	-2.22	+0.67
Acetic acid	-3.1	-3.97	-0.87

CHAPTER 4

RESULTS AND DISCUSSION

4.1 VCell Simulation

The Virtual Cell simulations successfully reproduced the diffusion of selected rice-derived bioactive compounds through a multilayer skin model consisting of the vehicle, lipid layer, corneocyte layer, viable epidermis, and dermis. The model generated time-dependent concentration profiles that illustrated how each compound moved across the different skin compartments.

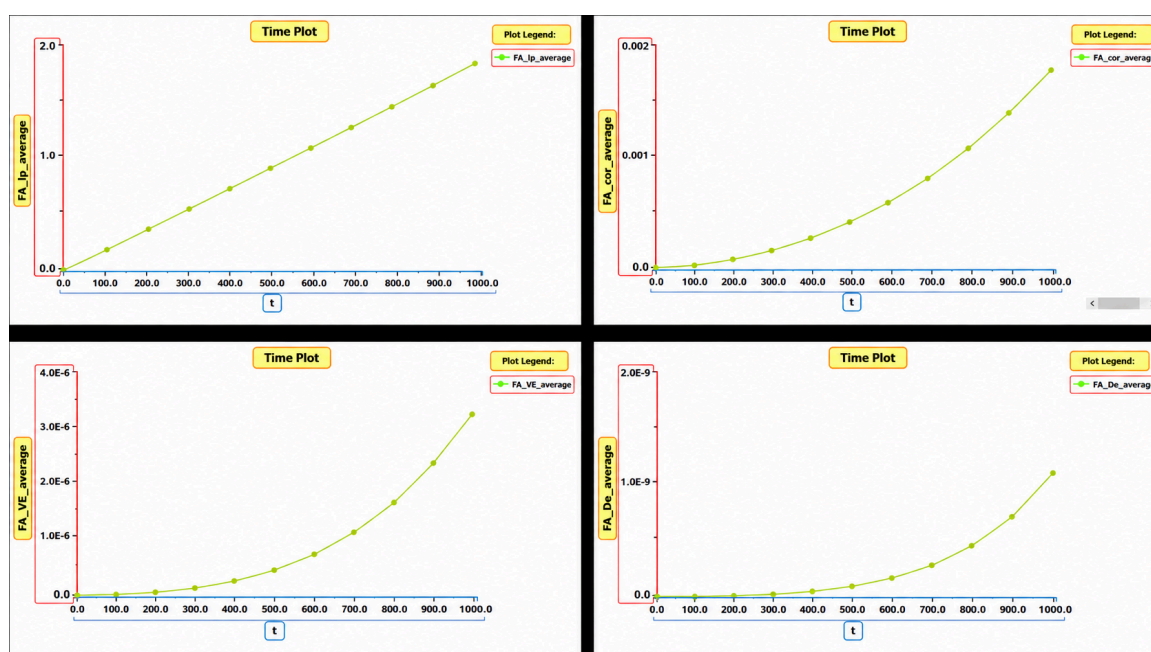


Figure 5: Time-dependent concentration profiles of compounds across different skin layers in the VCell model.

The simulation results showed a clear concentration gradient, with the highest concentration in the vehicle layer immediately after application. Over time, the compounds gradually diffused into the stratum corneum and then into the deeper layers of the skin. As expected, the concentration decreased with increasing depth, reflecting the barrier function of the skin. Differences in diffusion behavior were observed among the compounds. Molecules with more favorable physicochemical properties, particularly lower molecular weight and less negative logKp values, penetrated more efficiently and reached deeper skin layers within the

simulation period. In contrast, compounds with lower predicted permeability showed slower diffusion and remained concentrated in the upper layers.

The spatial concentration plots and time-course graphs provided detailed information on penetration depth, diffusion rate, and accumulation patterns. These outputs made it possible to compare the dermal transport behavior of different rice bioactives and identify compounds with stronger skin permeation potential.

Table 3: Predicted logKp values of selected rice-derived bioactive compounds obtained from VCell analysis.

Bioactive compound	VCell(logkp)
Ferulic acid	-3.99826
Caffeic acid	-4.147725254
p-Coumaric acid	-4.07477
Sinapic / Sinapinic acid	-4.12181
Protocatechuic acid	-4.123387539
γ-Oryzanol	-2.50553
Vanillic acid	-4.08772
Chlorogenic acid	-4.43333
Gallic acid	-4.18799
Phytic acid	-4.1414
Anthocyanins	-3.530958994
Cyanidin-3-O-glucoside	-4.45777
Cyanidin-3-O-galactoside	-4.45759
Pelargonidin-3-O-glucoside	-4.11558
Proanthocyanidins	-4.452465379
Proanthocyanidin dimers	-4.312925819
Proanthocyanidin trimers	-4.432813494
Catechin	-4.213578019
Epicatechin	-4.228962219
Tricin	-3.993785219
Luteolin	-3.986739694
Apigenin	-3.906505019
Rutin	-4.686672094
Quercetin	-4.12607

Kaempferol	-4.026928714
Myricetin	-4.205505619
Glutathione	-4.489198754
Citric acid	-4.32167

Overall, the VCell simulations demonstrated that computational modeling can effectively predict the movement of rice-derived compounds through the skin. The results complemented the machine learning analysis and provided a mechanistic understanding of how molecular properties influence dermal diffusion.

4.2 Machine Learning Analysis

The machine learning workflow developed in this study was able to estimate the permeability behavior of rice-derived bioactive compounds using their molecular properties. Descriptors related to molecular size, lipophilicity, polarity, hydrogen bonding, flexibility, and solubility were included during prediction analysis. The obtained results indicated that these physicochemical properties play a major role in determining the movement of compounds across skin layers.

The descriptor analysis revealed noticeable trends between molecular properties and permeability behavior. Compounds with smaller molecular size and suitable lipophilic characteristics generally showed better predicted diffusion tendencies. In contrast, compounds with higher polarity and larger structural size displayed lower permeability behavior, indicating stronger resistance to movement through the skin barrier.

Table 4: Comparison of regression model performance.

S. No.	Model	R ² Score	MAE	RMSE
1	Linear Regression	0.9323	0.1957	0.2433
2	Random Forest Regression	0.9639	0.1062	0.1777

3	Support Vector Regression (SVR)	0.9493	0.1127	0.2104
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Several regression approaches were explored during model development, including Random Forest Regression, Gradient Boosting, and Support Vector Regression. Among these methods, ensemble-based models produced more reliable predictions because they were able to learn complex relationships present within the descriptor data.

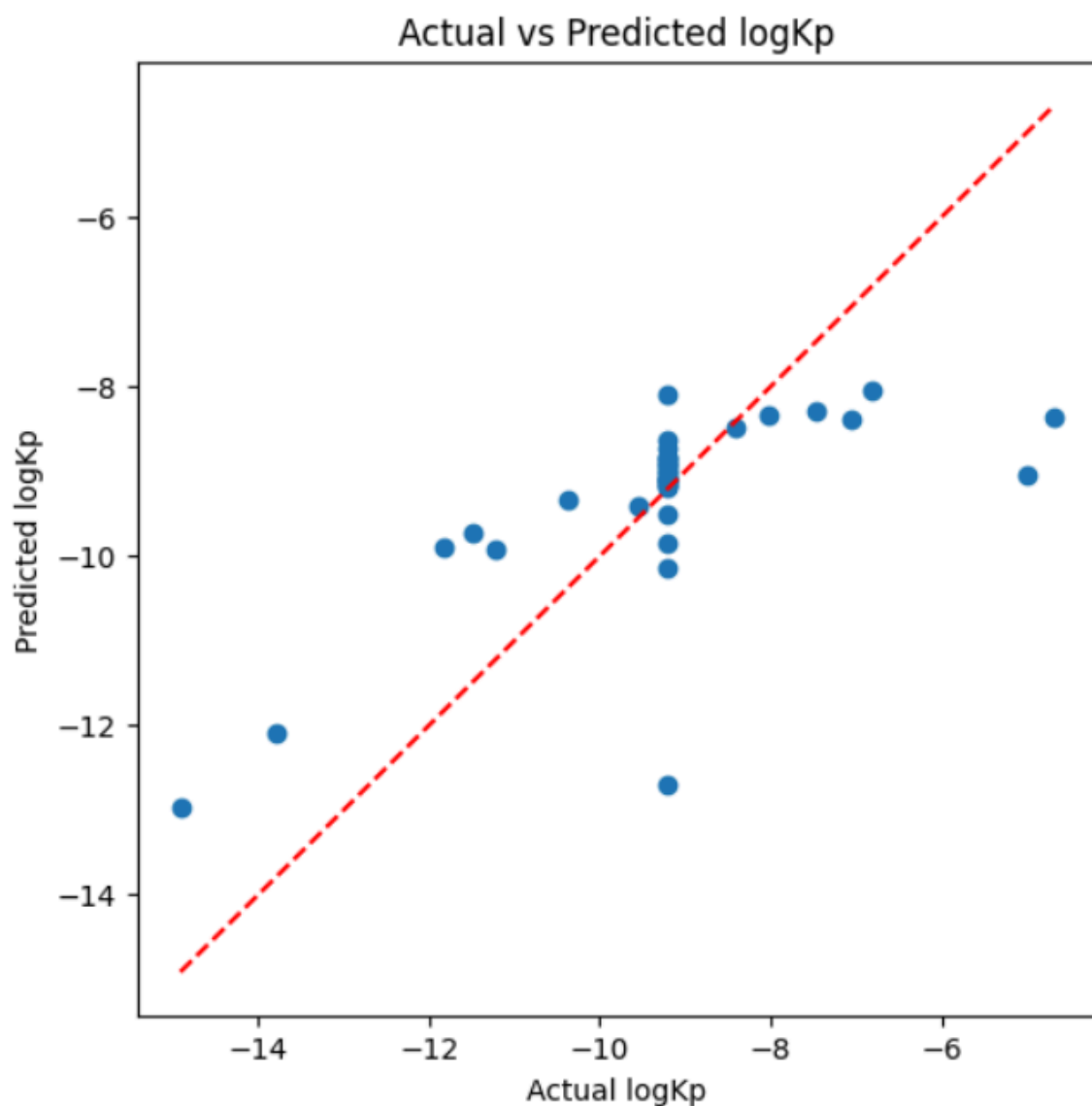


Figure 6: Actual vs predicted logKp values obtained from the regression model.

The comparison between observed and predicted permeability values showed close agreement, suggesting that the developed models performed effectively during prediction analysis.

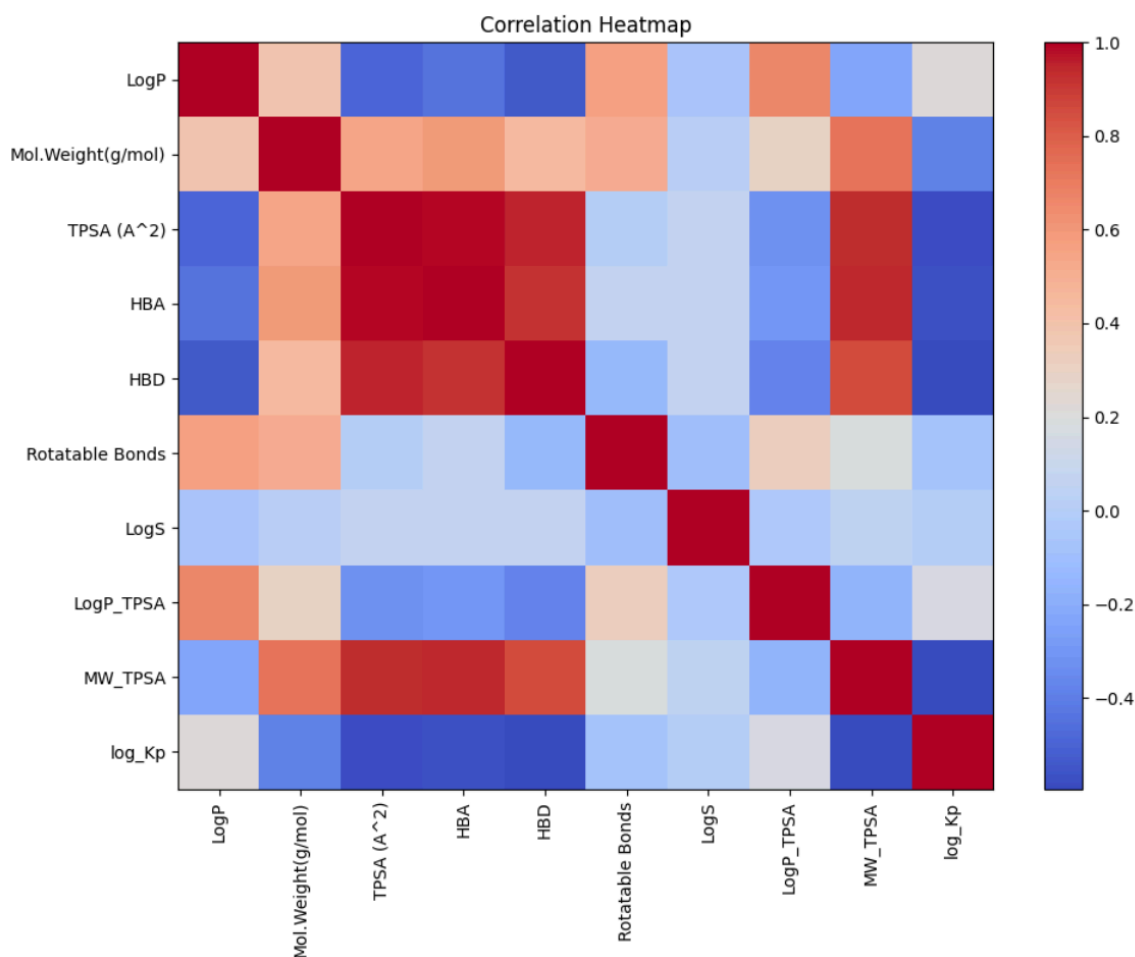


Figure 7: Correlation heatmap of molecular descriptors used in the study.

The cross-validation study demonstrated that the prediction models maintained relatively stable performance across different portions of the dataset. Descriptor contribution analysis further highlighted the importance of molecular weight, LogP, and TPSA in controlling permeability behavior.

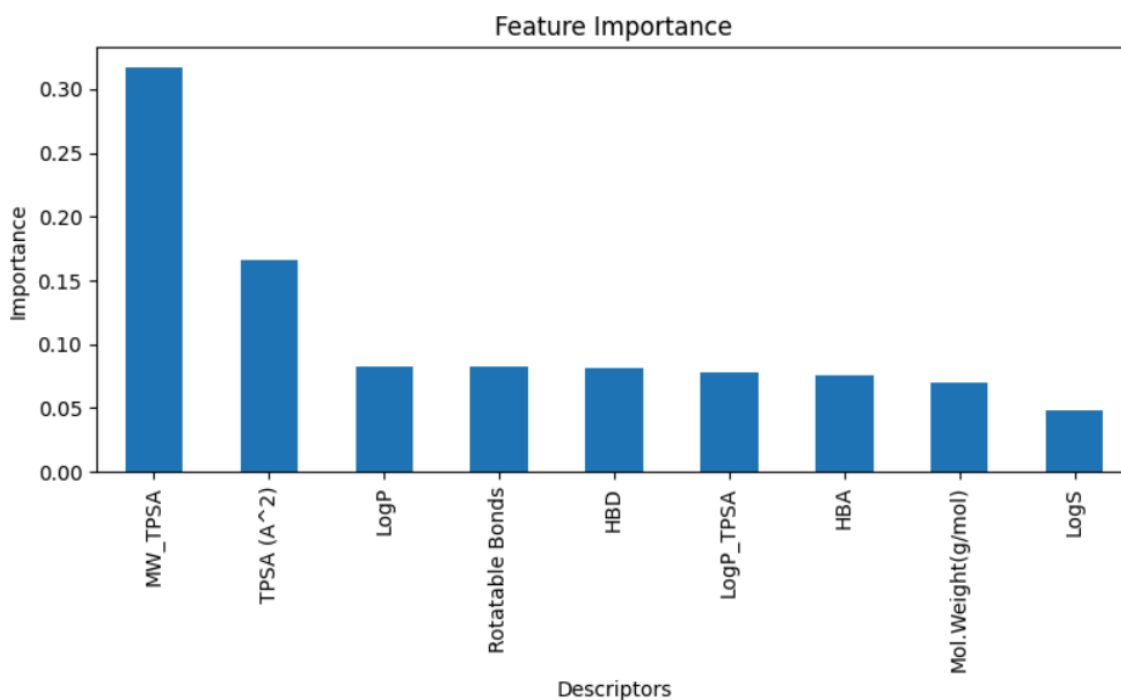


Figure 8: Feature importance of molecular descriptors.

These descriptors consistently showed strong influence on the predicted diffusion characteristics of the compounds.

Prediction-based ranking of the compounds helped identify rice bioactives with comparatively higher permeability potential. Compounds with balanced molecular properties generally showed improved transport tendencies, whereas highly polar or bulky compounds demonstrated weaker penetration behavior. Additional sensitivity analysis supported the observation that moderate lipophilic behavior contributes positively to dermal diffusion.

Overall, the machine learning study demonstrated that computational prediction methods can provide useful insight into the dermal absorption behavior of rice-derived compounds. The findings also supported the diffusion patterns observed in the VCell simulations and helped explain how molecular descriptors influence skin permeability characteristics.

CHAPTER 5

CONCLUSION

This study developed a computational approach to evaluate the skin permeability behavior of rice-derived bioactive compounds for topical and cosmetic applications. Molecular descriptors such as molecular weight, LogP, TPSA, hydrogen bonding properties, and solubility were analyzed to study their influence on dermal diffusion behavior.

Machine learning models including Linear Regression, Random Forest Regression, Gradient Boosting, and Support Vector Regression were applied for permeability prediction. The results showed that molecular size, lipophilicity, and polarity strongly influence skin permeability, while ensemble-based models provided more reliable prediction performance.

VCell simulations further demonstrated the diffusion behavior of compounds across different skin layers and highlighted the barrier role of the stratum corneum. The combined use of machine learning and simulation-based analysis provided an efficient and cost-effective framework for studying dermal transport behavior of rice-derived compounds.

Overall, the study showed that computational methods can effectively support the screening and evaluation of natural bioactive compounds for cosmetic and topical application

5.1 Limitations

- The study used a relatively small dataset of around 130 rice-derived bioactive compounds, which may limit model robustness and generalizability.
- Predictions were mainly based on computational simulations and molecular descriptors rather than direct experimental measurements.
- The VCell model employed a simplified skin representation and did not account for real human skin variability such as thickness, hydration, lipid composition, and metabolism.

Future Work

- Perform experimental validation through ex vivo skin permeation studies using human or animal skin to verify predicted permeability values.
- Expand the dataset with more natural and synthetic compounds to improve chemical diversity, model robustness, and prediction accuracy.
- Explore advanced machine learning and deep learning approaches such as Artificial Neural Networks, Graph Neural Networks, and ensemble boosting methods.
- Integrate molecular docking studies to investigate interactions between bioactive compounds and skin-related proteins or lipids.
- Apply the improved framework for the development and optimization of rice-derived compounds in cosmetic, topical, and therapeutic applications.